

Setup: $\underline{X} = (X_1, \dots, X_n)$ sample from dis^k with density
(pmf/pdf) $f(x|\theta)$; $\theta \in \Theta \subseteq \mathbb{R}^k$. Goal to learn θ

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, 1)$, To est. μ , $T_1(\underline{X}) = \frac{X_1 + \dots + X_n}{n}$
 $T_2(\underline{X}) = \frac{X_1 + X_2}{2}$

Not using full "information" if use T_2
Data $\underline{X}$ contain full information about μ , we should use all that info

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} U(0, \theta)$ $T_1(\underline{X}) = 2\bar{X} \xrightarrow{P} \theta$
 $T_2(\underline{X}) = \frac{n+1}{n} X_{(n)} \xrightarrow{P} \theta$

Check: T_1, T_2 unbiased

Principle of Data Reduction: Sufficient Statistics

$(X_1, \dots, X_n)$ have pmf/pdf $f(x|\theta)$; $\theta \in \Theta$

$\mathcal{X}$ = sample space = set of all possible values of $\underline{X}$

Any statistic $T(\underline{X})$ leads to "condensation/reduction of data" and it partitions $\mathcal{X}$

Ex. $X_1, X_2, X_3 \stackrel{iid}{\sim} \text{Ber}(p)$ $T(\underline{X}) = X_1 + X_2 + X_3$

(x_1, x_2, x_3)	T
(0, 0, 0)	0
(1, 0, 0), (0, 1, 0), (0, 0, 1)	1
(1, 1, 0), (1, 0, 1), (0, 1, 1)	2
(1, 1, 1)	3

Data Reduction: If we know $\underline{x}$, we can compute $T(\underline{x})$

But if we only know $T(\underline{x})$, we may not know $\underline{x}$

Any stat. $T(\underline{X})$ leads to data reduction. In the process we may lose some info about θ , that were available in full data $\underline{X}$

A sufficient stat $T(\underline{X})$ is a very special one, that reduces data, but still contains full info about θ , that were in $\underline{X}$

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bern}(p)$

$T(X) = \sum X_i$ (or $\bar{X}$) should be sufficient for estimating p

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$

$T(X) = \sum X_i$ (or $\bar{X}$) should be sufficient for estimating θ

$$H^{n \times n} = \begin{bmatrix} \frac{1}{\sqrt{n}} & \dots & \frac{1}{\sqrt{n}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \dots 0 \\ \vdots & & \\ \frac{1}{\sqrt{n(n-1)}} & \frac{1}{\sqrt{n(n-1)}} & \dots & \frac{-(n-1)}{\sqrt{n(n-1)}} \end{bmatrix}$$

$$\begin{aligned} \underline{X} &\sim N_n(\underline{\mu}, I_n) \\ \underline{Y} = H\underline{X} &\sim N_n(H\underline{\mu}, I) \\ &= (Y_1, \dots, Y_n) \end{aligned}$$

$$\text{Indep} \begin{cases} Y_1 = \sqrt{n}\bar{X} \sim N(\sqrt{n}\mu, 1) \\ Y_2 \sim N(0, 1) \\ \vdots \\ Y_n \sim N(0, 1) \end{cases} \quad \left\{ \begin{array}{l} \text{redundant} \end{array} \right.$$

No need to record all $X_1, \dots, X_n$. Just need $Y_1 = \sqrt{n}\bar{X}$ to estimate μ

Defⁿ: Let $\underline{X} = (X_1, \dots, X_n)$ have density (pmf/pdf) $f(\underline{x}|\theta)$, $\theta \in \Theta$
A statistic $T(\underline{X})$ is called sufficient statistic for θ

[OR for the family $\{f(x|\theta): \theta \in \Theta\}$] iff the conditional distn of $\underline{X}$ given $T(\underline{X})=t$ does not depend on $\theta \in \Theta$

1. If we know $T(\underline{X})=t$, no additional info is left in $\underline{X}$
2. If we know $T(\underline{X})=t$, we can generate a new obs $\underline{X}'$ that has same distn. as $\underline{X}$ (given t)

$$\underline{X}' | T=t \stackrel{d}{=} \underline{X} | T=t$$

$T(\underline{X}) = \underline{X}$ is always sufficient for θ

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bern}(\theta)$ Claim: $T(\underline{X}) = \sum_{i=1}^n X_i$ suff for θ
 $\sim \text{Bin}(n, \theta)$

Check

$$\mathbb{P}_{\theta}(X_1=x_1, \dots, X_n=x_n | T=t)$$

$$= \frac{\mathbb{P}(X_1=x_1, \dots, X_n=x_n, \sum X_i=t)}{\mathbb{P}(\sum X_i=t)}$$

$$= \frac{\mathbb{P}_{\theta}(X_1=x_1, \dots, X_n=x_n)}{\mathbb{P}_{\theta}(\sum_{i=1}^n x_i = t)} \mathbb{1}\left(\sum_{i=1}^n x_i = t\right)$$

$$= \frac{\theta^{\sum x_i} (1-\theta)^{n-\sum x_i}}{\binom{n}{t} \theta^t (1-\theta)^t} \mathbb{1}\left(\sum_{i=1}^n x_i = t\right)$$

$$= \frac{1}{\binom{n}{t}} \mathbb{1}\left(\sum_{i=1}^n x_i = t\right) \rightarrow \text{free of } \theta$$

$\therefore \sum X_i$ is sufficient for θ

Likelihood Principle: If $x, y \in \mathcal{X}$ s.t. $\mathcal{L}(\theta|x) \propto \mathcal{L}(\theta|y)$

Then inference (about θ) should be same based on x and y

Sufficiency Principle: If $x, y \in \mathcal{X}$ s.t. $T(x) = T(y)$ (where $T(X)$ is a sufficient stat for θ), then inference about θ should be same based on x and y

Ex. $X_1, X_2 \stackrel{iid}{\sim} \text{Bern}(\theta)$

Show that $T(X) = X_1 - X_2$ not suff for θ

$$\begin{aligned} & \mathbb{P}_\theta(X_1=0, X_2=0 | X_1 - X_2 = 0) \\ &= \frac{\mathbb{P}(X_1=0, X_2=0)}{\mathbb{P}(X_1=0, X_2=0) + \mathbb{P}(X_1=1, X_2=0)} \\ &= \frac{(1-\theta)^2}{(1-\theta)^2 + \theta^2} \text{ not free of } \theta \end{aligned}$$

Ex. $X \sim \mathcal{N}(0, \theta)$; $\theta > 0$ $T = |X|$

$$\mathbb{P}(X=t | |X|=t) = \mathbb{P}(X=-t | |X|=t) = \frac{1}{2} \text{ free of } \theta$$

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poi}(\lambda)$ $T(X) = \sum X_i$

$$\begin{aligned} \mathbb{P}_\lambda(X_1=x_1, \dots, X_n=x_n | T=t) &= \frac{\mathbb{P}_\lambda(X_1=x_1, \dots, X_n=x_n, T=t)}{\mathbb{P}_\lambda(T=t)} \\ &= \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod x_i!} \bigg/ \frac{t!}{e^{-n\lambda} (n\lambda)^t} \mathbb{1}_{(T=t)} \\ &= \frac{t!}{\prod x_i! n^t} \mathbb{1}_{(T=t)} \end{aligned}$$

H.W. $T(\underline{x}) = X_1 + \dots + X_{n-1} + 2X_n$. Is it suff for λ ?

$X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poi}(\lambda)$

Given any pmf/pdf how do we identify a sufficient stat

For discrete $(\underline{X})$:
$$\begin{aligned} P_{\theta}(\underline{X} = \underline{x}) &= P_{\theta}(\underline{X} = \underline{x}, T(\underline{X}) = T(\underline{x})) \\ &= \underbrace{P_{\theta}(\underline{X} = \underline{x} | T(\underline{X}) = T(\underline{x}))}_{\text{free of } \theta \text{ if } T \text{ suff stat}} P_{\theta}(T(\underline{X}) = T(\underline{x})) \end{aligned}$$

$$= h(\underline{x}) g_{\theta}(T(\underline{x}))$$

Neyman-Fisher Factorisation Theorem (NFFT)

Suppose $\underline{X} = (X_1, \dots, X_n)$ has joint pmf/pdf $f(\underline{x}|\theta); \theta \in \Theta$

$T(\underline{x})$ is a sufficient statistic for θ (or for the family $\{f(\underline{x}|\theta): \theta \in \Theta\}$)

iff $\exists$ functions h and g such that $\boxed{f(\underline{x}|\theta) = h(\underline{x}) g(T(\underline{x})|\theta)}$
where h depends only on $\underline{x}$ (free of θ) and g depends on both θ and $T(\underline{x})$

ex. $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poi}(\theta)$

$$f(\underline{x}|\lambda) = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod_{i=1}^n x_i!}$$

$\Rightarrow T(\underline{X}) = \sum_{i=1}^n X_i$ is a suff
stat for λ

$$= \underbrace{\left(\frac{1}{\prod_{i=1}^n x_i!} \right)}_{h(\underline{x})} \underbrace{\left(e^{-n\lambda} \lambda^{\sum x_i} \right)}_{g(\sum x_i | \theta)}$$

Pf: (for discrete X^*)

$(\Rightarrow)$ already proved

$(\Leftarrow)$ $\exists$ some $h(\cdot), g(\cdot, \cdot)$ s.t. $f(x|\theta) = h(x)g(T(x)|\theta)$

T.s. $T(X)$ is suff

$$\begin{aligned} & P_{\theta}[X = \underline{x} | T(X) = t] \\ &= \frac{P_{\theta}(X = \underline{x}, T(X) = t)}{P_{\theta}(T(X) = t)} \mathbb{1}_{(T(\underline{x}) = t)} \\ &= \frac{P_{\theta}(X = \underline{x})}{P_{\theta}(T(X) = t)} \mathbb{1}_{(T(\underline{x}) = t)} \\ &= \frac{P_{\theta}(X = \underline{x})}{\sum_{\underline{y}: T(\underline{y}) = t} P_{\theta}(X = \underline{y})} \mathbb{1}_{(T(\underline{x}) = t)} \\ &= \frac{h(\underline{x}) g(T(\underline{x})|\theta)}{\sum_{\underline{y}: T(\underline{y}) = t} h(\underline{y}) g(T(\underline{y})|\theta)} \mathbb{1}_{(T(\underline{x}) = t)} \\ &= \frac{h(\underline{x}) g(t|\theta)}{\left(\sum_{\underline{y}: T(\underline{y}) = t} h(\underline{y}) \right) g(t|\theta)} \mathbb{1}_{(T(\underline{x}) = t)} \\ &= \frac{h(\underline{x})}{\sum_{\underline{y}: T(\underline{y}) = t} h(\underline{y})} \mathbb{1}_{(T(\underline{x}) = t)} \text{ free of } \theta \quad \square \end{aligned}$$

Ex. Let $X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$ $\theta = (\mu, \sigma^2)$

$$\begin{aligned} f(\underline{x}|\mu, \sigma^2) &= \frac{1}{\sigma^n \sqrt{2\pi}^n} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2} \\ &= \frac{1}{\sigma^n \sqrt{2\pi}^n} e^{-\frac{1}{2\sigma^2} \left(\sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 \right)} \\ &= \underbrace{\frac{1}{\sqrt{2\pi}^n}}_{h(\underline{x})} \left(\underbrace{\frac{1}{\sigma^n} e^{-\frac{1}{2\sigma^2} \{ (n-1)S^2 + n(\bar{x} - \mu)^2 \}}}_{g(T(\underline{x})|\theta)} \right) \end{aligned}$$

$T_1(\underline{x}) = (\bar{x}, S^2)$ is suff for θ

$T_2(\underline{x}) = (\bar{x}, \sum_{i=1}^n (x_i - \bar{x})^2)$ is suff for θ

$T_3(\underline{x}) = (\bar{x}, \sum_{i=1}^n x_i^2)$ is suff for θ

By NFFT, $T = (\bar{x}, S^2)$ is (jointly) suff for $\theta = (\mu, \sigma^2)$

$\Rightarrow \bar{x}$ is suff for μ (when σ^2 unknown)

or S^2 is " " " " (μ " ")

NFFT holds when $\underline{X}$ is cont. r.v. (Pf. beyond scope for this class, uses Radon-Nikodym Derivative)

Both $T(\underline{x})$ and θ can be multidimensional.

Ex. Let $X_1, \dots, X_2 \stackrel{iid}{\sim} \mathcal{N}(\theta, \theta^2)$ $\theta \in \mathbb{R}$

Can there be a 1-dimensional sufficient stat. Think about it.